

Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	1 Mark

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The machine learning model is trained on a limited dataset, which may not represent all real-world applicant profiles.	Prediction accuracy may reduce when processing unseen or highly diverse applicant data.	High
2	The application currently does not store user prediction history or maintain a centralized database.	Previous predictions cannot be retrieved or analyzed for future reference.	Medium
3	The system is designed as a prototype for educational purposes and has not been integrated with real-time banking systems or external APIs.	Manual data entry is required, limiting its use in real-world banking environments.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	The application is hosted on a local Flask server and is suitable for a limited number of users.	Deploy the application on a cloud platform (e.g., Render, AWS, Azure) with load balancing and scalable server infrastructure to support more concurrent users.
2	Data Storage	Applicant data is read from a CSV dataset, and prediction history is not stored permanently.	Integrate a relational database such as MySQL or PostgreSQL to securely store applicant details, prediction history, and reports.

3	Performance	The system provides fast predictions for individual requests but is optimized for moderate workloads.	Optimize model loading, implement caching, and use a production-ready server (Gunicorn/Nginx) to improve response time and handle higher traffic efficiently.
4	Security	Basic input validation is implemented, but advanced security features are limited.	Add user authentication and authorization, encrypt sensitive data, enforce HTTPS, and implement secure API access with proper validation and logging.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on a cloud platform (e.g., Render, AWS, or Azure) for online access.	Within 1–2 months	Enables users to access the application from anywhere and improves availability.
Phase 2	Integrate a relational database (MySQL/PostgreSQL) to store applicant information and prediction history.	Within 3–4 months	Improves data management, enables record retrieval, and supports reporting and analytics.
Phase 3	Enhance the machine learning model using a larger, more diverse dataset and advanced algorithms for improved accuracy.	Within 5–6 months	Increases prediction accuracy, reliability, and overall system performance.
Phase 4	Integrate the system with banking APIs and implement advanced security features such as role-based authentication and data encryption.	Within 6–12 months	Makes the application suitable for real-world banking environments with enhanced security and automation.